

LECTURE 12 · DISCRETE MATHEMATICS

Relations select pairs

Which connections are allowed?



Prove a set law one element at a time

Fix an arbitrary element $x \in U$.

Complement of intersection

$$x \in (A \cap B)^c \iff \neg(x \in A \wedge x \in B)$$

Use De Morgan

$$\iff x \notin A \vee x \notin B$$

Translate back

$$\iff x \in A^c \cup B^c$$

$$(A \cap B)^c = A^c \cup B^c$$

Set laws mirror logic

Law	Union	Intersection
Idempotent	$A \cup A = A$	$A \cap A = A$
Commutative	$A \cup B = B \cup A$	$A \cap B = B \cap A$
Associative	$(A \cup B) \cup C = A \cup (B \cup C)$	$(A \cap B) \cap C = A \cap (B \cap C)$
Identity	$A \cup \emptyset = A$	$A \cap U = A$
Domination	$A \cup U = U$	$A \cap \emptyset = \emptyset$

Reference: read $\cup$ as OR and $\cap$ as AND.

Negation swaps union and intersection

Law	Identity
De Morgan	$(A \cup B)^c = A^c \cap B^c ; (A \cap B)^c = A^c \cup B^c$
Complement	$A \cup A^c = U ; A \cap A^c = \emptyset$
Double complement	$(A^c)^c = A ; U^c = \emptyset ; \emptyset^c = U$
Absorption	$A \cup (A \cap B) = A ; A \cap (A \cup B) = A$

Distribute either operation over the other

AND over OR

$$\begin{aligned} A \cap (B \cup C) \\ = (A \cap B) \cup (A \cap C) \end{aligned}$$

OR over AND

$$\begin{aligned} A \cup (B \cap C) \\ = (A \cup B) \cap (A \cup C) \end{aligned}$$

Membership gives the familiar distributive laws of logic.

Pairs remember order. Sets do not.

Set

$$\{a, b\} = \{b, a\}$$

Ordered pair

$$(a, b) \neq (b, a) \quad \text{if } a \neq b$$

A tuple records positions: first, second, third, ...

ORDERED PAIRS

A Cartesian product lists every possible pair

$$A = \{a, b\}, \quad B = \{1, 2, 3\}.$$

$$A \times B = \{(x, y) : x \in A \wedge y \in B\}$$

	1	2	3
a	$(a, 1)$	$(a, 2)$	$(a, 3)$
b	$(b, 1)$	$(b, 2)$	$(b, 3)$

$$|A \times B| = |A| |B| = 2 \cdot 3 = 6$$

Reversing a pair changes its meaning

$$A = \{a, b\}, \quad B = \{1, 2, 3\}.$$

01 Give a pair in $A \times B$ but not in $B \times A$.

$(a, 1) \in A \times B$, but $(a, 1) \notin B \times A$.

02 Can $A \times B$ and $B \times A$ ever be equal?

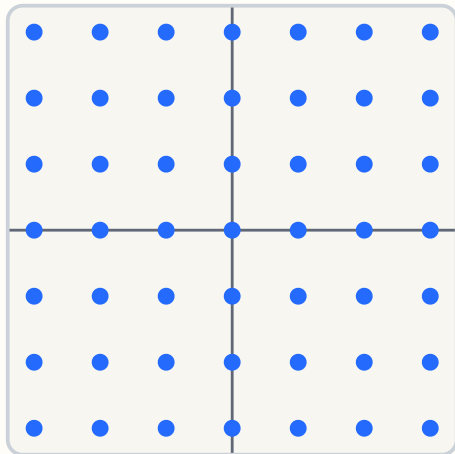
Yes, exactly when $A = B$ or either set is empty.

EXERCISE 2 · TRY FIRST

What does each product look like?

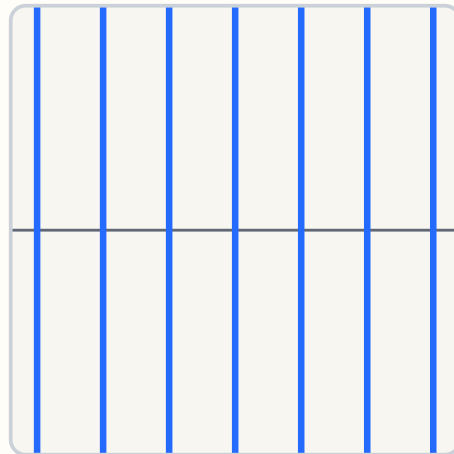
Each product is a set of points (x, y) . Sketch it before revealing.

01 $\mathbb{Z} \times \mathbb{Z}$



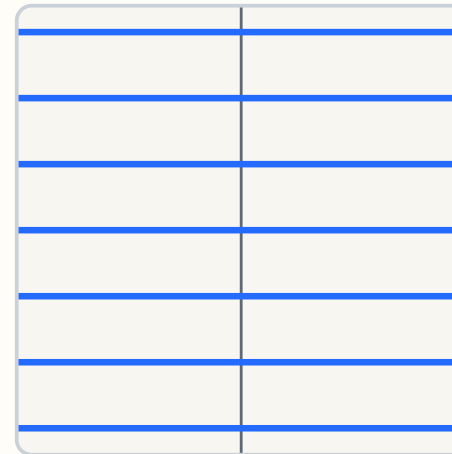
Integer points only.

02 $\mathbb{Z} \times \mathbb{R}$



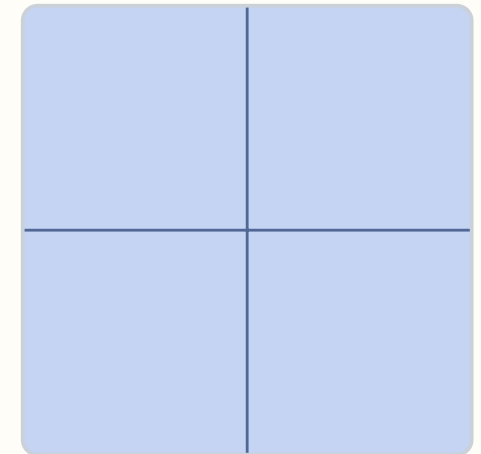
Vertical lines at integer x .

03 $\mathbb{R} \times \mathbb{Z}$



Horizontal lines: not $\mathbb{Z} \times \mathbb{R}$.

04 $\mathbb{R} \times \mathbb{R}$



The whole plane, $\mathbb{R}^2$.

The first factor limits x . The second factor limits y .

EXERCISE 3 · TRY FIRST

Can a disk be a product?

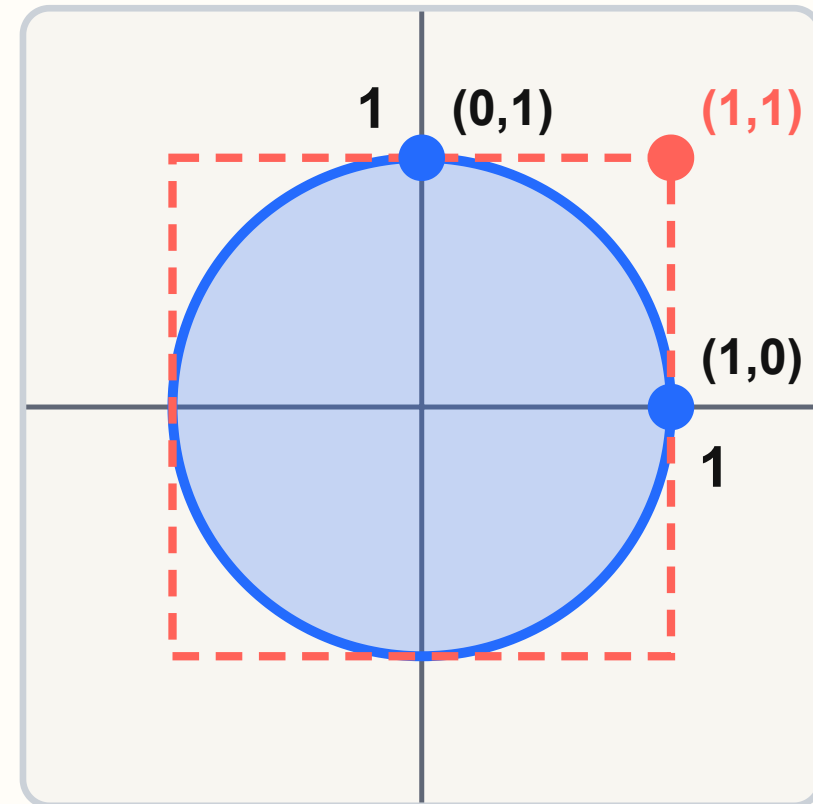
$$D = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}.$$

Find $P, Q \subseteq \mathbb{R}$ with $D = P \times Q$, or show that none exist.

$(1, 0), (0, 1) \in D$ force $1 \in P$ and $1 \in Q$.

Then $(1, 1) \in P \times Q$, but $1^2 + 1^2 = 2 > 1$.

So no such P and Q exist.



A rule can select any subset of $\mathbb{R} \times \mathbb{R}$, not just products.

Six possible pairs. Three allowed.

Robots $A = \{a, b\}$. Docks $B = \{1, 2, 3\}$.

a can use docks 1 and 3. b can use dock 2.

Start from all $2 \cdot 3 = 6$ pairs in $A \times B$.

Keep three: $R = \{(a, 1), (a, 3), (b, 2)\}$.

The product lists possibilities. The relation selects connections.

A relation is a set of ordered pairs

$$R \subseteq A \times B$$

Read the notation

$$aRb \iff (a, b) \in R$$

One selection

$$R = \{(2, p), (3, q), (4, r)\}$$

$$A = \{1, 2, 3, 4\}, B = \{p, q, r\}$$

First entry from A. Second entry from B.

A relation may select any subset

No pairs

$$R = \emptyset$$

Every candidate is excluded.

Every pair

$$R = A \times B$$

Every candidate is included.

One pair is allowed too. So are several pairs with the same first entry.

EXERCISE 4 · TRY FIRST

How many docking rules are possible?

$|A| = 2, \quad |B| = 3$. A rule may allow any subset of $A \times B$.

01 How many candidate pairs?

$$2 \cdot 3 = 6.$$

02 Each pair is either allowed or excluded. How many relations?

$$2^6 = 64.$$

For finite sets: $2^{|A||B|}$ relations from A to B.

A relation “on A ” uses A twice

$$R \subseteq A \times A, \quad A = \{0, 1, 2, 3\}.$$

Rule	Selected pairs
$=$	$\{(0, 0), (1, 1), (2, 2), (3, 3)\}$
$<$	$\{(0, 1), (0, 2), (0, 3), (1, 2), (1, 3), (2, 3)\}$
$\geq$	All (a, b) with $a \geq b$: 10 pairs.

The underlying set is part of the definition.

EXERCISE 5 · TRY FIRST

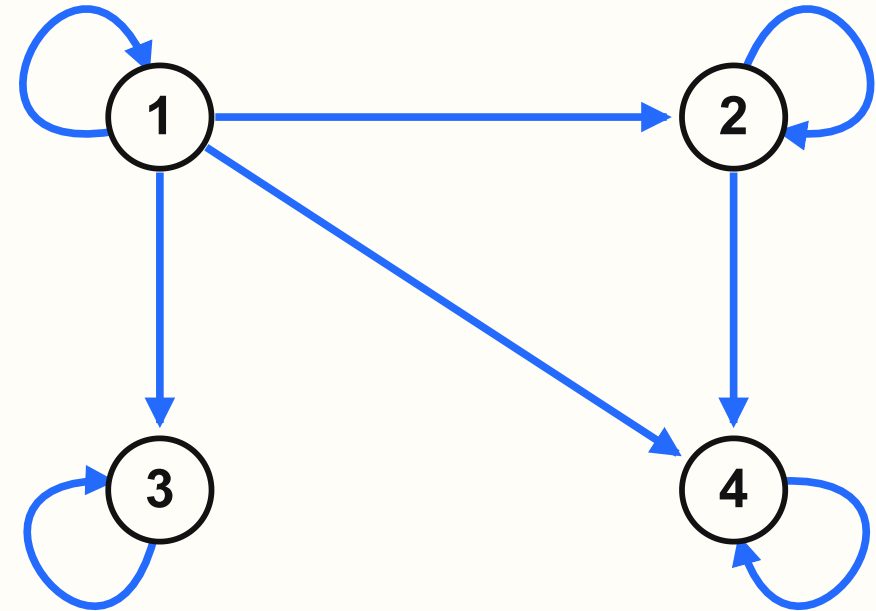
Which numbers divide which?

$A = \{1, 2, 3, 4\}$. List all pairs.

$$aRb \iff a \mid b$$

$1 \rightarrow 1, 2, 3, 4$

$2 \rightarrow 2, 4; \quad 3 \rightarrow 3; \quad 4 \rightarrow 4$



$a \mid b$ means $b = ak$ for some integer k .

A rule gives a pair its meaning

Objects	Keep the pair when...	Example
City $\rightarrow$ country	City is the capital.	(Riyadh, Saudi Arabia)
Integers	$a \mid b$	(2, 6)
Real $\rightarrow$ integer	$ x - y \leq 1$	(2.4, 2), (2.4, 3)

Always state the sets and the rule.

Infinite relations can have short rules

All six relations are on $\mathbb{Z}$.

Relation	Keep (a,b) when...	Relation	Keep (a,b) when...
R_1	$a < b$	R_4	$a = b \vee a = -b$
R_2	$a > b$	R_5	$a = b + 1$
R_3	$a = b$	R_6	$a + b \leq 3$

To test a pair, substitute its two entries.

EXERCISE 6 · TRY FIRST

Which rules accept each pair?

$R_1 : a < b$; $R_2 : a > b$; $R_3 : a = b$;

$R_4 : a = b \vee a = -b$; $R_5 : a = b + 1$; $R_6 : a + b \leq 3$.

01 (1, 1)

R_3, R_4, R_6

02 (1, 2)

R_1, R_6

03 (2, 1)

R_2, R_5, R_6

04 (1, -1)

R_2, R_4, R_6

05 (2, 2)

R_3, R_4

Relations can use triples too

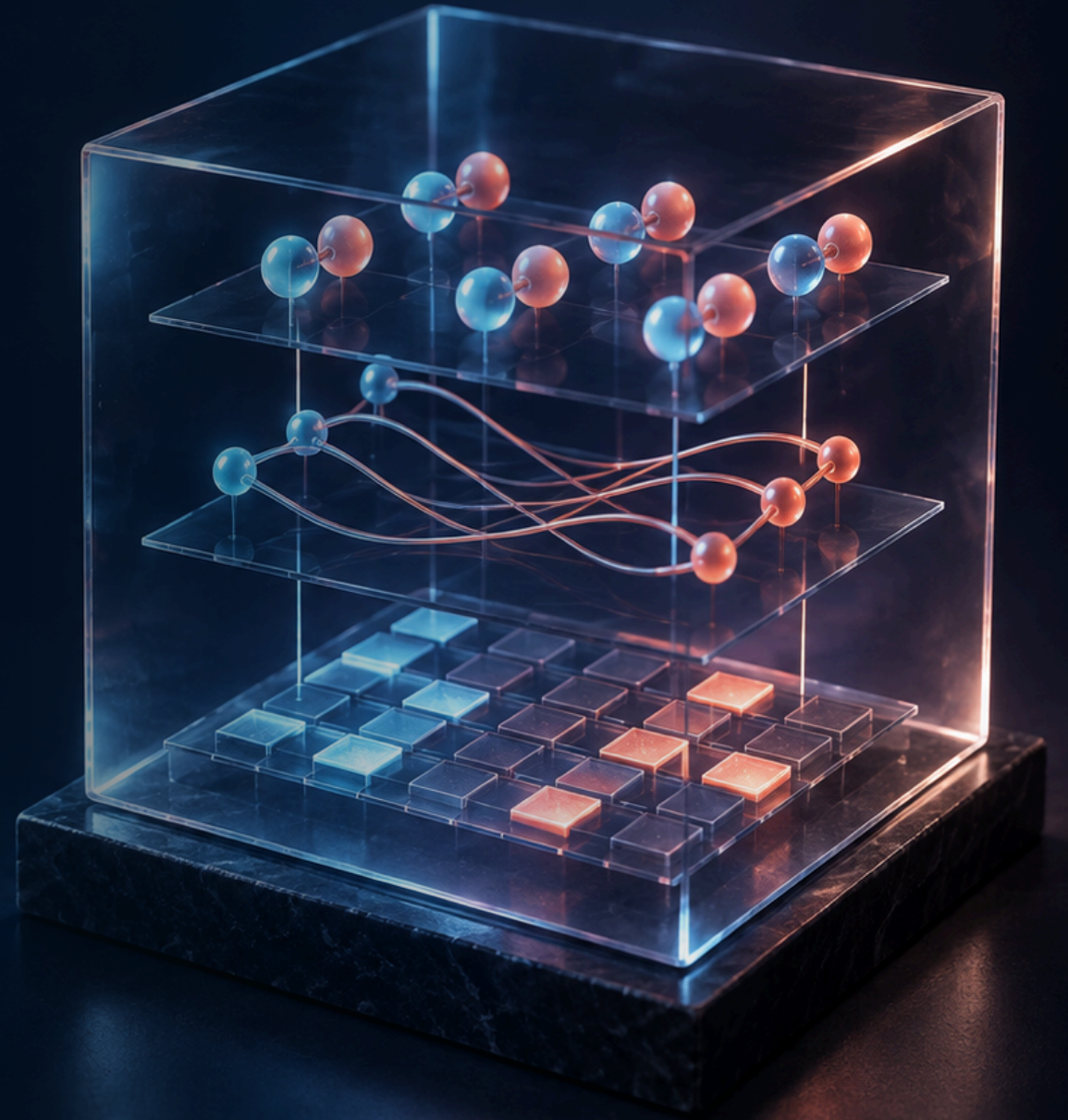
Rule	Candidates	Shape
$x = y$	$\mathbb{R} \times \mathbb{R}$	A line
$x^2 + y^2 + z^2 \leq 1$	$\mathbb{R} \times \mathbb{R} \times \mathbb{R}$	A solid ball

Binary uses pairs. Ternary uses triples.

ACT II · READ THE PAIRS

One relation. Four views.

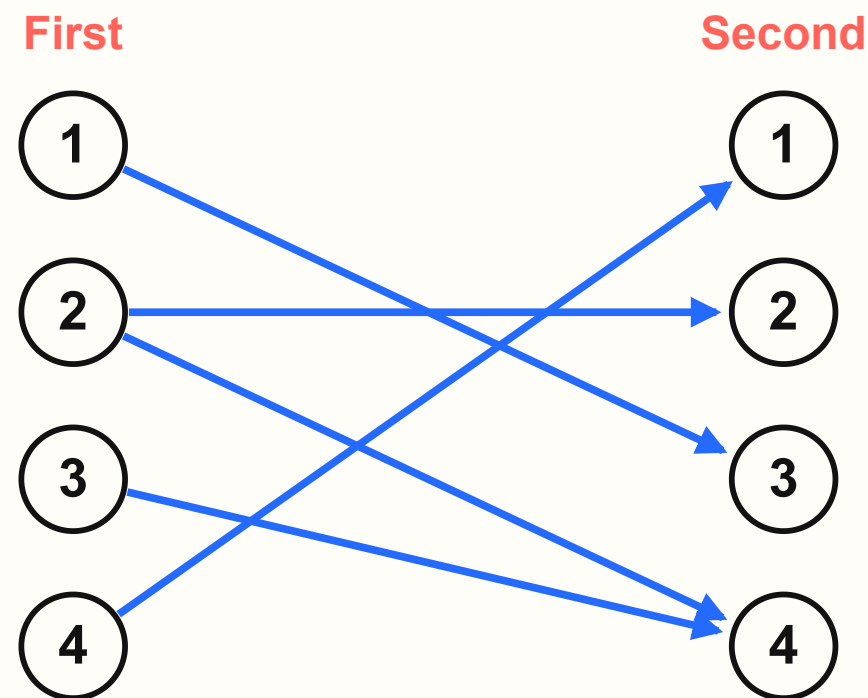
List. Arrows. Matrix. Plot.



An arrow records one ordered pair

$A = \{1, 2, 3, 4\}$. Read left to right.

$R = \{(1, 3), (2, 2), (2, 4), (3, 4), (4, 1)\}$



Two copies of A keep the two positions separate.

A matrix records every yes and no

Row = first entry.

Column = second entry.

$$M_{ab} = 1 \iff (a, b) \in R$$

	1	2	3	4
1	0	0	1	0
2	0	1	0	1
3	0	0	0	1
4	1	0	0	0

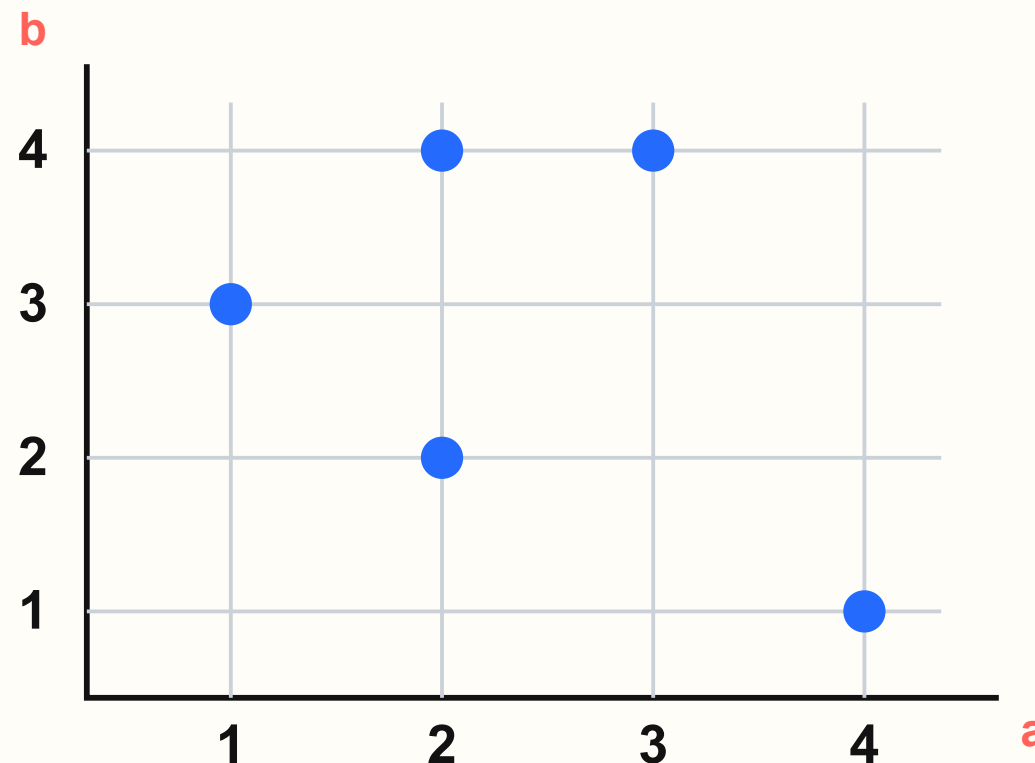
Same five pairs. Five 1s.

RELATIONS

A coordinate plot marks the same pairs

Horizontal = first entry.
Vertical = second entry.

$$(2, 4) \in R$$



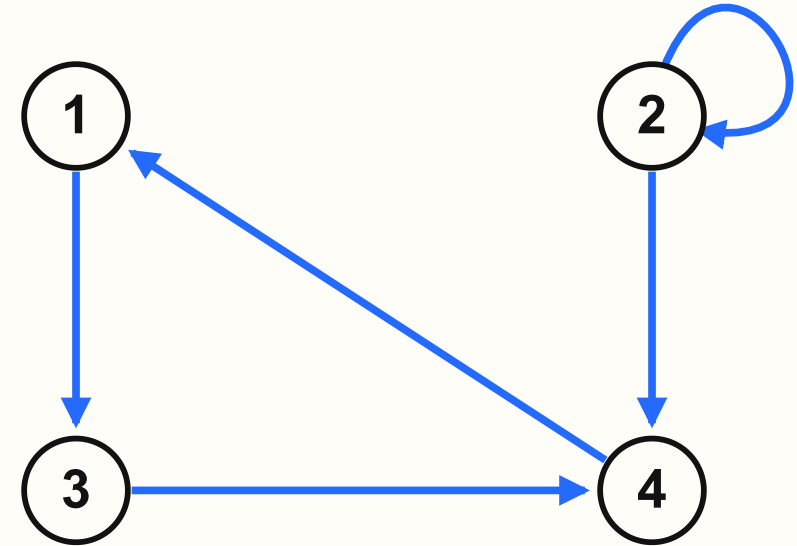
A matrix row runs downward. A plot's second axis runs upward.

RELATIONS

A digraph uses each element once

One vertex per element.
One arrow per pair.

$(2, 2)$ **becomes a loop.**



Pairs, arrows, 1s, and points carry the same information.

EXERCISE 7 · TRY FIRST

Decode the docking matrix

Rows are robots a, b . Columns are docks 1, 2, 3.

01 List the allowed pairs.

$$R = \{(a, 1), (a, 3), (b, 2)\}.$$

02 Which docks can robot a use?

Docks 1 and 3.

	1	2	3
a	1	0	1
b	0	1	0

Read the row first.

The inverse reverses every pair

$$R^{-1} = \{(b, a) : (a, b) \in R\}$$

Original

$$R = \{(x, y), (y, z), (y, x)\}$$

Inverse

$$R^{-1} = \{(y, x), (z, y), (x, y)\}$$

$(R^{-1})^{-1} = R$. Reverse twice; recover the original.

ACT III · TEST THE RULES

Patterns become properties

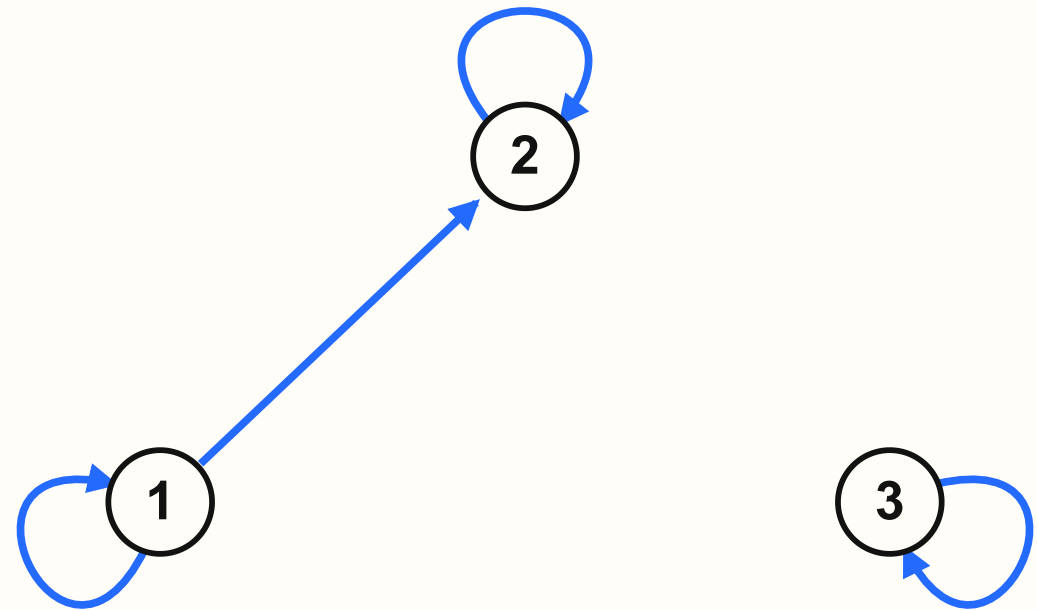
Loops. Reversals. Shortcuts.



Reflexive: every element has a loop

For every $x \in A$:

$$xRx$$



Other arrows are unrestricted.

No loops is stronger than a missing loop

Irreflexive

$$\forall x \in A : (x, x) \notin R$$

No element has a loop.

Not reflexive

$$\exists x \in A : (x, x) \notin R$$

At least one loop is missing.

On $A = \{1, 2\}$, $R = \{(1, 1)\}$ is neither reflexive nor irreflexive.

EXERCISE 8 · TRY FIRST

Find the missing loops

On $A = \{1, 2, 3, 4\}$: which relations are reflexive?

01 $R_1 = \{(1, 1), (2, 2), (3, 3), (4, 4)\}$

Yes.

02 $R_2 = \{(1, 1), (1, 4), (2, 2), (3, 3), (4, 2)\}$

No: $(4, 4)$ is missing.

03 $R_3 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$

Yes.

04 $R_4 = \{(1, 3), (2, 2), (3, 1), (2, 4), (3, 3), (4, 4)\}$

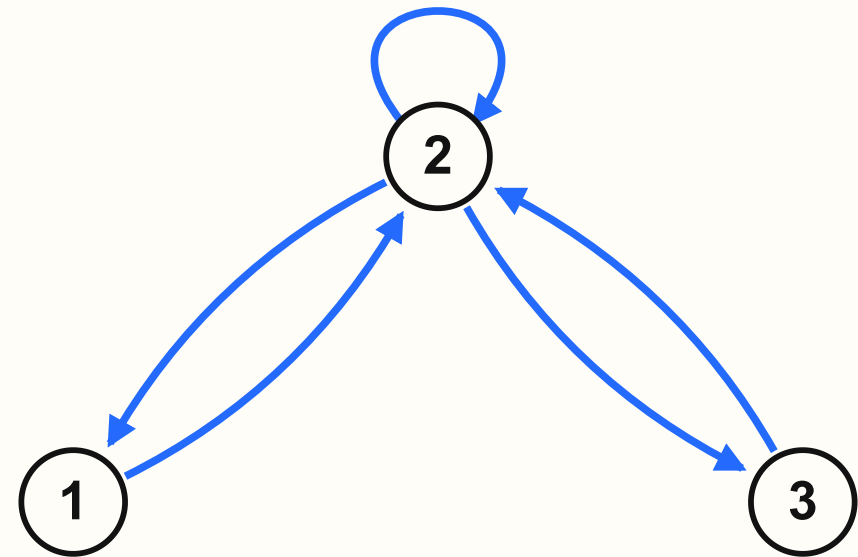
No: $(1, 1)$ is missing.

Name one missing pair for each “no.”

Symmetric: every arrow has a reverse

For all $x, y \in A$:

$$xRy \implies yRx$$



Equivalently, $R = R^{-1}$. Loops reverse themselves.

Find the missing reverse

On $A = \{1, 2, 3, 4\}$: which relations are symmetric?

01 $R_1 = \{(1, 1), (2, 2), (3, 3), (4, 4)\}$

Yes.

02 $R_2 = \{(1, 1), (1, 4), (2, 2), (3, 3), (4, 2)\}$

No: $(1, 4) \in R_2$, but $(4, 1) \notin R_2$.

03 $R_3 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$

Yes.

04 $R_4 = \{(1, 3), (2, 2), (3, 1), (2, 4), (3, 3), (4, 4)\}$

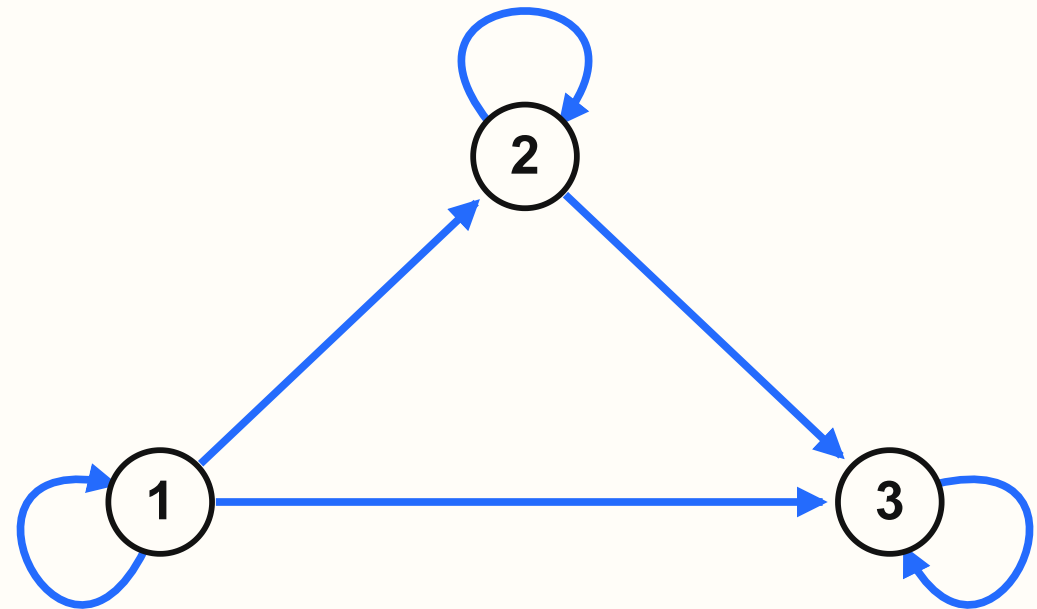
No: $(2, 4) \in R_4$, but $(4, 2) \notin R_4$.

A single unpaired arrow is enough.

Antisymmetric: two-way pairs must be loops

For all $x, y \in A$:

$$xRy \wedge yRx \implies x = y$$



$\leq$ is antisymmetric. Loops are allowed.

Asymmetric also forbids loops

Antisymmetric

$$xRy \wedge yRx \implies x = y$$

$\leq$: two-way pairs only at one element.

Asymmetric

$$xRy \implies \neg(yRx)$$

$<$: no two-way pairs, including loops.

Asymmetric = antisymmetric + irreflexive.

EXERCISE 10 · TRY FIRST

Symmetric, antisymmetric, both, or neither?

All relations are on $A = \{1, 2, 3\}$.

01 $\{(1, 1), (2, 2), (3, 3)\}$

Both. Only loops.

02 $\{(1, 2)\}$

Antisymmetric only.

03 $\{(1, 2), (2, 1), (2, 3)\}$

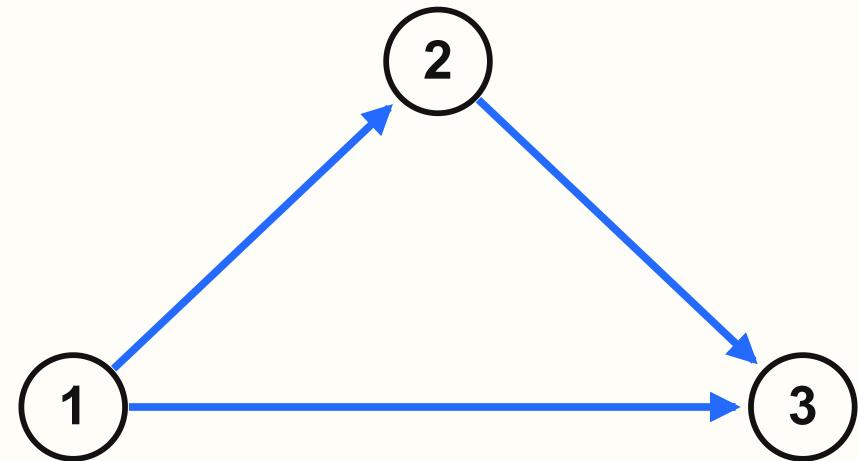
Neither: $1 \leftrightarrow 2$; no $3 \rightarrow 2$.

The two properties are separate tests.

Transitive: every two-step route has a shortcut

For all $x, y, z \in A$:

$$xRy \wedge yRz \implies xRz$$



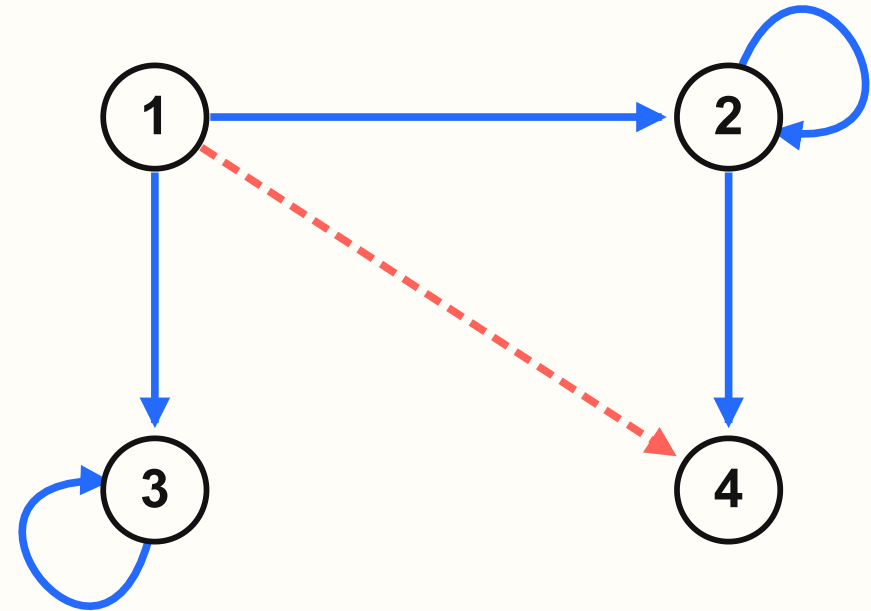
The shortcut must already belong to R.

EXERCISE 11 · TRY FIRST

Find one missing shortcut

$A = \{1, 2, 3, 4\}$. Is the pictured relation transitive?

No: $1R2$ and $2R4$, but $\neg(1R4)$.



Dashed coral arrow appears with the answer.

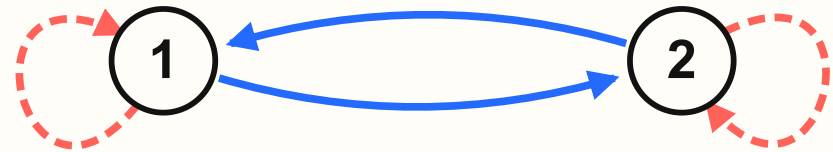
A round trip demands a loop

$$R = \{(1, 2), (2, 1)\} \text{ on } A = \{1, 2\}.$$

Transitive?

No: $1R2$ and $2R1$ require $1R1$.

The reverse route also requires $2R2$.



The first and third elements may be equal.

Antitransitive forbids every shortcut

Antitransitive

$$xRy \wedge yRz \implies \neg(xRz)$$

On $\mathbb{Z}$: $y = x + 1$.

Not transitive

$$\exists x, y, z : xRy \wedge yRz \wedge \neg(xRz)$$

At least one required shortcut is missing.

Successor steps: +1, then +1, gives +2; never +1.

EXERCISE 12 · TRY FIRST

Is “nearby” transitive?

Toy audio matcher: real-valued pitch scores x and y .

$$xRy \iff |x - y| \leq 1$$

01 Reflexive?

Yes: $|x - x| = 0 \leq 1$.

02 Symmetric?

Yes: $|x - y| = |y - x|$.

03 Transitive? Give a counterexample.

No: $0R0.75$ and $0.75R1.5$, but $\neg(0R1.5)$.

Two small gaps can make one large gap.